Problem Chosen	University Physics Competition	Team Control Number
A		758

Controlled Descent and Propellant Estimation for a First-Stage Rocket Booster

Summary

We model first-stage recovery with coupled downrange, vertical, and pitch dynamics, using a reference liquid-oxygen and methane propulsion design. A nominal one-ignition trajectory satisfies the selected touchdown criteria with a separation propellant inventory of approximately 26.23 t. Fixed-schedule perturbation cases motivate further guidance analysis. Planning separately for known initial-state and steady-wind variations gives eleven feasible cases at a common 30 t inventory and fixed ship position. The remaining assessment includes uncertain measurements, disturbances arising during flight, actuator response, and structural and thermal limits.

Draft plan: Complete the summary with the selected full-flight strategy, separation propellant load, recovery site, and principal safety limits. Keep the final summary within 300 words.

Contents

1 Introduction **4**
1.1 Background . . . 4
1.2 Restatement of the Problem . . . 4

2 Assumptions **4**

3 Notations **5**

4 Model Force Analysis **5**
4.1 Gravity and Standard Atmosphere . . . 5
4.2 Axial and Crossflow Aerodynamics . . . 6
4.3 Reference Propulsion Parameters . . . 7

5 Model Motion Analysis **7**
5.1 Momentum Balance and Pitch Dynamics . . . 7
5.2 Initial Conditions and Contact Geometry . . . 8
5.3 Analytic Baselines and Scale Estimates . . . 8
5.4 Numerical Integration and Local Powered-Braking Check . . . 9

6 Landing Strategy and Propellant Estimation **9**
6.1 Search and minimum-propellant formulation . . . 9
6.2 Nominal strategy and separation budget . . . 10
6.3 Dependence on the separation direction . . . 12

7 Sensitivity and Uncertainty Analysis **13**
7.1 Initial Flight-Path Angle and Atmospheric Coverage . . . 13
7.2 Wind, Initial Attitude, and Fixed-Ship Capture . . . 13
7.3 Separation-Time Replanning at a Fixed Inventory . . . 14
7.4 Error in the Wind Used for Planning . . . 15
7.5 Tracking with Control Margins . . . 15
7.6 Flight Loads and Aerothermal Exposure . . . 16
7.7 Whole-Flight Parameters and Reserve . . . 18

8 Model Evaluation **20**
8.1 Overview and Comparison of Model Levels . . . 20
8.2 Verification and Validation . . . 20
8.3 P1 Internal Verification and Numerical Convergence . . . 20
8.4 Strengths and Weaknesses . . . 21
8.5 Discussion of Sources of Error . . . 21

9 Further Discussion **21**
9.1 Attitude Stability and Vertical Touchdown . . . 21
9.2 Structural Loads and Aerothermal Constraints . . . 21
9.3 Ship Placement, Wind, and Deck Motion . . . 22
9.4 Conclusions and Practical Recommendations . . . 22

A Reproducibility and Source Code **23**

B Supplementary Derivations and Results **23**

Report on the Use of AI **23**

1 Introduction

1.1 Background

Reusable first-stage boosters require controlled descent and a safe vertical landing before refurbishment and reuse.

1.2 Restatement of the Problem

At separation, the booster has an altitude of 70 km and a speed of 6000 km/h. Its dry mass is 130000 kg, its height is 65 m, and its diameter is 7.0 m. It uses liquid oxygen and methane. The recovery ship may be placed at a convenient location. The task is to analyze a safe descent and controlled vertical landing, and estimate the fuel required at separation [11]. The problem specifies the initial speed magnitude. The direction affects the time aloft, downrange distance, and state at landing-burn ignition, so it is an explicit scenario input. We use *propellant* for the combined mass of liquid oxygen and methane and *fuel* for the methane component.

2 Assumptions

The booster is initially approximated as a rigid circular cylinder. This external geometry does not establish a uniform internal mass distribution. Atmospheric drag is retained using air-relative velocity and altitude-dependent density. The baseline translational model assumes planar motion and zero wind; steady uniform wind is a subsequent scenario, rather than a justification for neglecting aerodynamic torque.

Aerodynamic moments depend on the offset between the center of pressure and center of mass. A prescribed-axis translational model therefore cannot establish attitude stability or vertical touchdown. Rotational dynamics and control authority must be assessed separately.

Earth rotation is provisionally omitted from the baseline and requires a quantitative sensitivity check. Its omission is not justified by freedom in ship placement: Coriolis acceleration can alter velocity and consequently aerodynamic loading. The initial flight-path angle and all unspecified engine and aerodynamic parameters remain explicit scenario inputs.

The deck is initially stationary and horizontal. Position denotes the booster center of mass; contact geometry must account for vehicle length and, when specified, landing gear and deck height.

Reference Scenario and Mission Conventions

The reference flight-path angle and body pitch at separation are both 15° above horizontal, with zero initial pitch rate. Alternate flight-path scenarios use 0° and 30°, with initial pitch matched to the respective flight-path angle. Ground-relative planar velocity is used. Wind is zero in the nominal case; deterministic horizontal wind scenarios are -10, 0, and $+10$ m/s. These scenario points are analysis choices, not statistical confidence bounds.

The ship is stationary, the deck is horizontal at sea level, and landing gear is omitted. A nominal ship position may be chosen separately for each nominal scenario. Perturbation tests for an already positioned ship retain that ship location. The selected touchdown

tolerances are 5 m horizontal position error, 1 m/s horizontal speed, 2 m/s vertical speed, 2° tilt from upright, and 1°/s pitch rate.

The initial inventory includes a minimum usable touchdown reserve of 5000 kg; unusable residual is assumed zero in the first model. The 80000 kg trial inventory is an initialization, not a fuel requirement. Consumed propellant, reserve, and any surplus are accounted for without double counting. At an oxygen-to-methane mass ratio of 3.5, methane represents 1/4.5 of total propellant mass.

At most three ignitions are allowed as a reference operational assumption. Ignition delay and minimum on/off durations are zero, and actuator rate limits are omitted. Throttle and gimbal changes are therefore idealized as instantaneous; this is not a manufacturer capability claim. No independent reaction-control system or active aerodynamic fins are assumed. Engine-off attitude motion is passive.

Sensitivity cases vary effective specific impulse over 300, 330, and 360 s; axial coefficient over 0.4, 0.8, and 1.2; crossflow coefficient over 0.8, 1.2, and 1.6; and pressure-center offset over $-0.05L$, 0, and $+0.05L$. These deterministic variations test the reference design assumptions. No verified structural or thermal allowable is available: peak dynamic pressure, load, and heating diagnostics must be reported, and touchdown feasibility alone does not establish structural or thermal safety.

3 Notations

Symbol	Meaning	Unit
$h, \mathbf{r}$	Altitude and position	m
$\mathbf{v}$	Velocity	m/s
m, m_d, m_p	Total, dry, and propellant mass	kg
T, I_{sp}	Thrust and specific impulse	N, s
ρ, C_D, A	Air density, drag coefficient, reference area	kg/m^3, 1, m^2
γ_0	Initial flight-path angle	rad
q	Dynamic pressure	Pa

4 Model Force Analysis

4.1 Gravity and Standard Atmosphere

We use a local downrange–vertical plane. Position refers to the center of mass (COM), and the pitch angle points from the tail toward the nose. Gravity acts through the COM and has magnitude

$$g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \qquad g_0 = 9.80665 \text{ m/s}^2, \quad R_E = 6356766 \text{ m}. \tag{1}$$

At 70 km this gives $g = 9.5942$ m/s^2, about 2.17% below g_0. This is a spherical, nonrotating reference approximation, not a latitude-resolved gravity model.

Atmospheric properties follow the lower layers of the 1976 standard atmosphere [3]. Geometric altitude h is converted to geopotential altitude $H = R_E h/(R_E + h)$. In a layer

with base (H_b, T_b, p_b) and temperature gradient λ,

$$T_a = T_b + \lambda(H - H_b), \tag{2}$$

$$p_a = \begin{cases} p_b(T_a/T_b)^{-g_0/(R_a\lambda)}, & \lambda \neq 0, \\ p_b \exp[-g_0(H - H_b)/(R_a T_b)], & \lambda = 0, \end{cases} \tag{3}$$

$$\rho = \frac{p_a}{R_a T_a}, \qquad R_a = 287.05287 \text{ J/(kg K)}. \tag{4}$$

These expressions follow from hydrostatic balance $dp_a/dH = -\rho g_0$ and the ideal-gas law. The implemented range ends at $H = 84.852$ km; trajectories outside that range are rejected rather than assigned extrapolated density.

4.2 Axial and Crossflow Aerodynamics

The cylindrical approximation gives $A_{end} = \pi D^2/4 = 38.4845$ m^2 and side area $DL = 455$ m^2. Define body directions

$$\mathbf{e} = (\cos\theta, \sin\theta), \quad \mathbf{n} = (-\sin\theta, \cos\theta), \quad \mathbf{u} = \mathbf{v} - \mathbf{w}, \quad u_a = \mathbf{u}\cdot\mathbf{e}, \quad u_n = \mathbf{u}\cdot\mathbf{n}. \tag{5}$$

Instead of applying one coefficient to a changing projected area, we separate axial and crossflow contributions. Cylinder research motivates treating crossflow as a distinct aerodynamic regime [2]; our constant coefficients are low-order engineering assumptions, not a fit to that report.

The axial force is

$$F_a = -\tfrac{1}{2}\rho C_a A_{end} u_a |u_a|. \tag{6}$$

For an equivalent crossflow strip at signed axial arm s, rotation adds the local normal velocity ωs. Thus

$$dF_n = -\tfrac{1}{2}\rho C_n D(u_n + \omega s)|u_n + \omega s|\, ds. \tag{7}$$

With N midpoint strips of length $\Delta s = L/N$ and $s_i = -L/2 + (i - 1/2)\Delta s + \Delta l$,

$$\mathbf{F}_D = F_a\mathbf{e} + \sum_i F_{n,i}\mathbf{n}, \qquad M_a = \sum_i s_i F_{n,i}. \tag{8}$$

The shifted equivalent distribution represents an uncertain pressure-center offset Δl; it is not a physical surface mesh. We evaluate density at the COM and neglect its variation across the 65 m body. Axial force acts along the centerline. Figure 1 defines the forces and signs.

For zero rotation, $M_a = \Delta l F_n$. Consequently, even spatially uniform wind can generate a moment when $\Delta l \neq 0$. For a centered distribution and pure rotation, direct integration gives

$$M_a = -\frac{\rho C_n D L^4}{64}\omega|\omega|. \tag{9}$$

This opposes rotation. More generally, the mechanical power transferred by the modeled air-relative forces satisfies

$$\mathbf{F}_D \cdot \mathbf{u} + M_a\omega = -\tfrac{1}{2}\rho C_a A_{end}|u_a|^3 - \tfrac{1}{2}\rho C_n D \sum_i \Delta s |u_n + \omega s_i|^3 \leq 0. \tag{10}$$

This identity provides a useful sign and implementation check. It concerns motion relative to air, not ground-frame energy in moving wind.

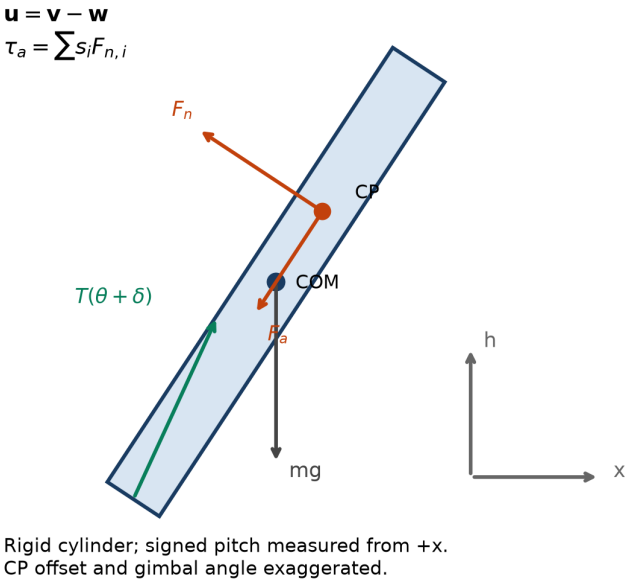


Figure 1: Planar reference geometry and force conventions. The pressure-center displacement is exaggerated. Positive gimbal angle at the tail produces a negative pitch moment.

4.3 Reference Propulsion Parameters

The maximum landing thrust is set to 2.846 MN by analogy with the published BE-4 thrust scale [1]. We select effective $I_{sp} = 330$ s, oxygen-to-methane mass ratio 3.5, minimum throttle 0.4, and gimbal limit 10° as reference-design assumptions. The mixture-ratio choice is informed by a methane/oxygen mission study [4]; that study's vacuum specific impulse is not assigned to this landing engine. We set $C_a = 0.8$, $C_n = 1.2$, and $\Delta l = 0.05L$. Parameter ranges are specified in the P0 reference contract and must be explored before a final fuel recommendation.

The standard effective-exhaust-velocity definition gives $c = I_{sp} g_0$ and

$$\dot{m} = -T/c, \qquad T = 0 \quad \text{or} \quad 0.4T_{max} \le T \le T_{max}. \tag{11}$$

This consumes both oxygen and methane. Pressure dependence and actuator transients are represented only through the effective parameter assumptions at this stage.

5 Model Motion Analysis

5.1 Momentum Balance and Pitch Dynamics

The state is $\mathbf{y} = (x, h, v_x, v_h, m, \theta, \omega)$. Accounting for exhaust momentum through thrust, the open-system translational balance is $m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{F}_D + \mathbf{F}_T$. One must not add a second

$\dot{m}\mathbf{v}$ term to this thrust-based equation. The coupled model is

$$\dot{x} = v_x, \qquad \dot{h} = v_h, \tag{12}$$

$$\dot{v}_x = \frac{F_{D,x} + T\cos(\theta+\delta)}{m}, \qquad \dot{v}_h = -g(h) + \frac{F_{D,h} + T\sin(\theta+\delta)}{m}, \tag{13}$$

$$\dot{m} = -\frac{T}{g_0 I_{sp}}, \qquad \dot{\theta} = \omega, \tag{14}$$

$$I_d\dot{\omega} = M_a - \frac{L}{2}T\sin\delta, \qquad I_d = \frac{m_d(3R_b^2 + L^2)}{12}. \tag{15}$$

The thrust application point is $-(L/2)\mathbf{e}$, so its cross product with thrust gives the negative sign above. The dry body is modeled as a uniform cylinder. Remaining propellant is idealized as concentrated at the COM with negligible pitch inertia and no additional rotational angular-momentum flux. Hence I_d is constant. A distributed-tank model would require consistent treatment of changing inertia and exhaust angular momentum; the present equation does not claim to describe those effects.

5.2 Initial Conditions and Contact Geometry

For each selected initial flight-path angle,

$$x_0 = 0, \quad h_0 = 70000 \text{ m}, \quad (v_{x,0}, v_{h,0}) = \frac{6000}{3.6}(\cos\gamma_0, \sin\gamma_0), \quad m_0 = m_d + m_{p,0}. \tag{16}$$

The reference uses $\gamma_0 = \theta_0 = 15^\circ$ and $\omega_0 = 0$. A trial $m_{p,0} = 80000$ kg initializes calculations; it is not a computed minimum.

For a cylindrical body with no landing gear, the vertical support extent below the COM is $(L/2)|\sin\theta| + R_b|\cos\theta|$. First deck contact therefore occurs when

$$f(\mathbf{y}) = h - h_{deck} - \frac{L}{2}|\sin\theta| - R_b|\cos\theta| = 0. \tag{17}$$

Only an upright body contacts at $h - h_{deck} = L/2$. Speed, attitude, pitch-rate, propellant reserve, and position relative to the chosen ship must be assessed separately. The local diagnostic below checks the former quantities, not fixed-ship position capture.

5.3 Analytic Baselines and Scale Estimates

For constant gravity, zero thrust, and zero drag,

$$h(t) = h_0 + v_{h,0}t - \tfrac{1}{2}g_0t^2, \qquad x(t) = v_{x,0}t. \tag{18}$$

At 15°, $(v_{x,0}, v_{h,0}) = (1609.88, 431.37)$ m/s. The constant-g_0 apex estimate is 79.49 km after 43.99 s. This is an analytic scale estimate, not the integrated reference trajectory. In the variable-gravity vertical vacuum limit, the conserved specific energy is

$$E = \tfrac{1}{2}v_h^2 - \frac{g_0R_E^2}{R_E + h}. \tag{19}$$

Without opposing external acceleration in the thrust direction, integrating $dv = -c\,dm/m$ yields

$$\Delta v = c\ln(m_0/m_f). \tag{20}$$

For 210 t initially and 135 t finally, this gives 1429.86 m/s. It is ideal propulsive capability, not a required landing velocity increment or proof of feasibility. Atmospheric drag may remove speed while gravity and steering alter the propulsion requirement.

With mass frozen, no drag, and vertical maximum thrust, a useful stopping-distance estimate is

$$d_{stop} \simeq \frac{v_h^2}{2(T_{max}/m - g_0)}, \qquad T_{max} > mg_0. \tag{21}$$

For 180 t and a 90 m/s descent, it is 674.5 m. Mass loss and drag are omitted from this estimate. The maximum-thrust weight ratio is 1.382 at 210 t and 2.232 at dry mass, showing why instantaneous mass must enter the dynamics.

5.4 Numerical Integration and Local Powered-Braking Check

A fixed-step fourth-order Runge–Kutta integrator advances the coupled equations. Step size is reduced near the deck, and contact is located by linear interpolation across a sign change of Eq. (17). This does not imply fourth-order event accuracy. A conservative propellant guard stops before a step could cross dry mass; it prevents negative inventory but does not return the exact depletion time.

To test the powered and rotational implementation independently of whole-flight optimization, we start a local case at $h = 1200$ m, $v_h = -90$ m/s, $v_x = 0$, and $m = 180000$ kg. Define height above upright contact $z = h - L/2$ and a descent-speed reference

$$v_d = -\min\{90, \sqrt{2k_z \max(z, 0.01) + v_c^2}\}, \quad k_z = 1.5 \text{ m/s}^2, \quad v_c = 1 \text{ m/s}. \tag{22}$$

A vertical acceleration command $a_c = k_v(v_d - v_h)$, $k_v = 4\ \text{s}^{-1}$, supplies the throttle request $[m(g_0 + a_c) - F_{D,h}]/T_{max}$, clipped to the permitted operating range. This vertical-axis test controller uses g_0 as an approximation and does not provide general two-dimensional guidance. Pitch control requests

$$M_{req} = I_d[k_p(\pi/2 - \theta) - k_d\omega] - M_a, \quad \delta = \arcsin\left[\text{clip}\left(-\frac{M_{req}}{(L/2)T}, -1, 1\right)\right], \tag{23}$$

where $k_p = 0.5\ \text{s}^{-2}$, $k_d = 1.2\ \text{s}^{-1}$, followed by gimbal clipping. With no saturation, this gives the stable error equation $\ddot{e} + k_d\dot{e} + k_p e = 0$. Saturation and engine-off motion invalidate that linear closed-loop conclusion and must be checked in the full trajectory.

At a 0.005 s step the nominal case contacts after 35.97505 s with $v_h = -1.50291$ m/s and consumes 22679.12 kg. A separate test with initial pitch 89° and initial rate 0.005 rad/s ends with approximately zero pitch error and $v_x = 0.4671$ m/s. Its horizontal drift is 16.56 m, exceeding the 5 m fixed-ship position tolerance if the ship remains at $x = 0$. Thus attitude recovery is demonstrated locally, but fixed-ship capture is not. Figure 2 shows these implementation checks; they do not estimate separation fuel demand.

6 Landing Strategy and Propellant Estimation

6.1 Search and minimum-propellant formulation

We first propagated the P1 equations with altitude-triggered ignition and a feedback law. During the burn, the pitch target blends a biased retrograde direction into an upright target with lateral-velocity feedback. The throttle follows a decreasing vertical-speed

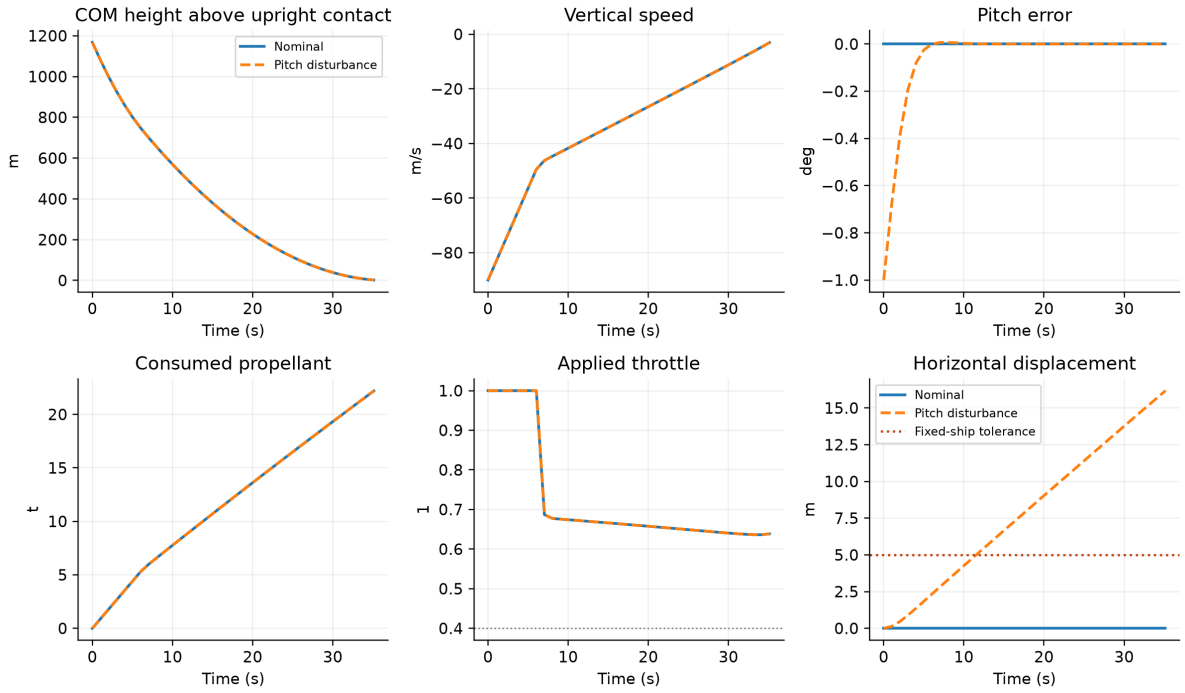


Figure 2: Local powered-braking verification, not a separation-to-landing solution. The disturbed case starts at 89° pitch with 0.005 rad/s pitch rate. The 5 m line highlights that position capture is not guaranteed by attitude recovery.

profile and the gimbal supplies a pitch-tracking moment. A search over ignition altitude, propellant inventory, pitch bias, and the vertical-speed profile furnished feasible seeds for one and two ignitions.

We then minimized $m(0) - m_d$ over the phase durations, throttle and gimbal commands, and trajectory states. The phase interfaces share all seven P1 states. Coast phases have zero thrust. Each powered phase obeys $0.4 \le u_T \le 1$ and $|\delta| \le 10°$. The separation state follows P0, with initial mass free. The endpoint uses cylindrical contact geometry, P0's deck-relative velocity and attitude tolerances, and a minimum 5 t usable reserve. The nominal ship position is a free downrange coordinate and is fixed at the selected position during replay. Interior target values of $v_x = 0$, $v_h = -1.5\,\mathrm{m/s}$, upright pitch and zero pitch rate, and a numerical reserve margin of 75 kg left room for discretization error.

We initialized Hermite–Simpson direct collocation from the feedback solutions [5]. We refined the candidates using RK4 multiple shooting with eight substeps per control interval and a local trust region in phase duration and pitch. CasADi supplied derivatives to IPOPT [6, 7]. We accepted a candidate only if replay with the original P1 code located the first deck contact and passed the full P0 touchdown checks at both 0.01 s and 0.005 s. Throttle and gimbal were interpolated separately within each phase. The calculation used the specified instantaneous actuator model.

6.2 Nominal strategy and separation budget

For the 15° reference separation in still air, the selected trajectory uses one ignition. The booster coasts for 176.68 s and ignites at a center-of-mass altitude of 5.876 km, while

moving downrange at 669 m/s and descending at 580 m/s. The landing burn lasts 30.79 s and reduces both velocity components while bringing the body upright. First deck contact occurs at 207.47 s, with the stationary ship placed 278.310 km downrange.

With that ship position held fixed in the replay, the contact point is within 1 mm of the target. The deck-relative horizontal and vertical velocities are -0.011 and $-1.533\,\mathrm{m/s}$, with magnitudes below the 1 and 2 m/s limits. The body is within 0.001° of upright and the pitch-rate magnitude is $0.059^\circ/\mathrm{s}$, below the respective limits of 2° and $1^\circ/\mathrm{s}$. The usable propellant remaining is 5.078 t, above the 5 t requirement. Together, these results establish touchdown feasibility for the reference case under the P0 criteria.

The separation inventory follows from the mass balance

$$m_{p,0} = m_{\mathrm{burned}} + m_{\mathrm{reserve}} + m_{\mathrm{residual}} + m_{\mathrm{surplus}}. \tag{24}$$

The planned usable reserve is 5 t, the unusable residual is set to zero, and surplus is usable propellant remaining above the planned reserve. In the one-ignition case, the coast consumes no propellant. The 21.1505 t burned during landing and the 5.0776 t retained at contact give a separation load of 26.2281 t. The 5 t reserve is already included in that load. With the assumed oxygen-to-methane mass ratio of 3.5, methane accounts for 1/4.5 of the load, or 5.828 t. Liquid oxygen accounts for the remaining 20.400 t.

Table 1: Nominal 15° strategies that satisfy the P0 touchdown criteria in P1 replay. Each strategy uses its own selected ship position. Masses are in metric tonnes and rounded independently.

Quantity	One ignition	Two ignitions
Separation propellant	26.228	27.607
Consumed propellant	21.151	22.516
Planned usable reserve	5.000	5.000
Unusable residual	0.000	0.000
Usable surplus at contact	0.078	0.091
Methane at separation	5.828	6.135
Liquid oxygen at separation	20.400	21.472
Peak dynamic pressure (kPa)	297.3	295.7
Peak proper acceleration (g_0)	7.70	7.70

The accepted two-ignition solution in Table 1 carries 1.38 t more than the one-ignition solution. Its first pulse lasts approximately 0.001 s, at the numerical duration floor, and burns only 0.36 kg. A separate feedback candidate with a 1.5 s entry burn also satisfies the touchdown criteria and requires 38.65 t. The one-ignition strategy therefore gives the lowest accepted inventory among the schedules found for the reference case. This comparison is limited to the candidates returned by the search and the assumed actuator model.

Rerunning the optimization with a nominal shooting interval of 0.55 s instead of 0.85 s reduces the selected inventory by about 70 kg. For the selected schedule held fixed, reducing the replay step from 0.01 s to 0.005 s changes the consumed mass by 0.72 kg. A further reduction to 0.0025 s changes it by 0.15 kg and preserves touchdown feasibility. The replay comparisons isolate the effect of integrating the same controls at different time steps.

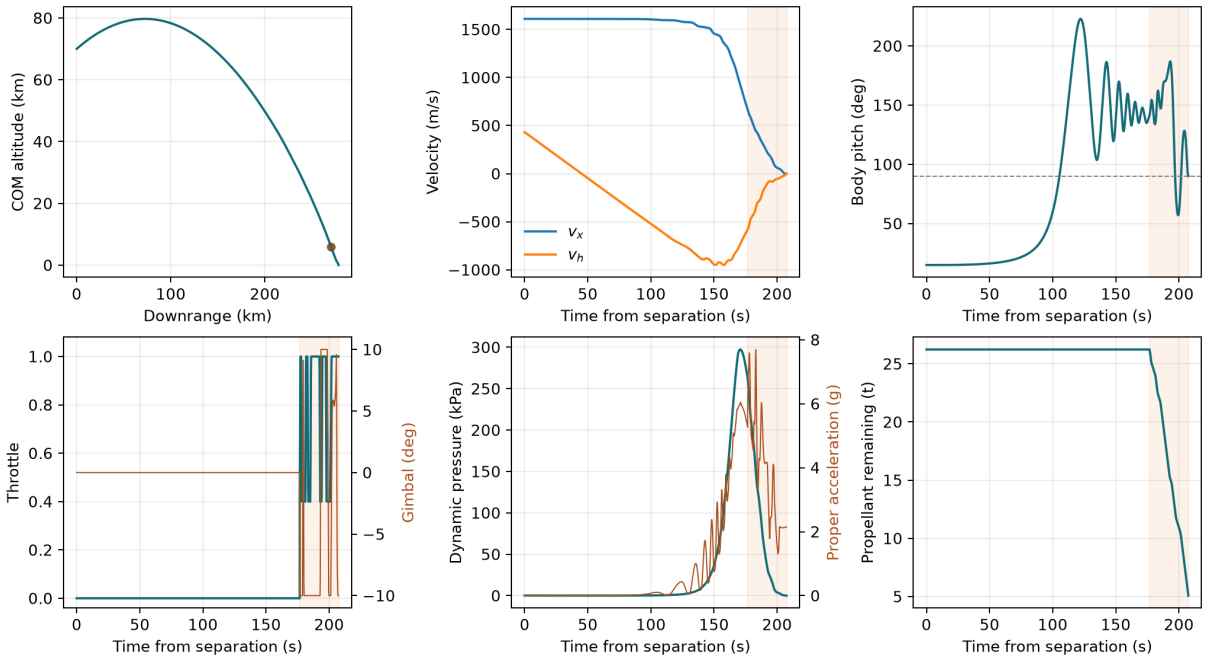


Figure 3: Selected one-ignition trajectory at 15°. The dot and shaded interval mark ignition and the landing burn, respectively. Dynamic pressure peaks before ignition, while proper acceleration peaks during the burn.

For the selected trajectory, dynamic pressure reaches 297.3 kPa at 9.69 km altitude, before ignition. Proper acceleration reaches $7.70 g_0$ at 2.89 km during the burn. These peaks describe the modeled flight environment. Assessing vehicle feasibility also requires structural limits, a heating model with thermal limits, and a model of actuator response during the large pitch excursions.

6.3 Dependence on the separation direction

The problem specifies the speed at separation but leaves its direction open. We compare initial flight-path angles of 0°, 15°, and 30° above horizontal, with the body pitch initially aligned with the flight path in each case. We use the same propulsion assumptions, touchdown criteria, and 5 t usable reserve requirement. For each angle we select a recovery site and hold that site fixed when replaying the candidate. Table 2 reports the lowest inventory found to satisfy the touchdown criteria for each case.

Table 2: Accepted strategies for the three separation angles. Ship position is measured downrange from the separation point. The separation load includes the planned 5 t reserve.

Angle	Powered phases	Ship position (km)	Separation load (t)
0°	Landing burn	196.00	24.71
15°	Landing burn	278.31	26.23
30°	Ascent braking and landing burn	260.00	67.00

The 0° and 15° solutions both coast to a single landing burn. The 15° trajectory first climbs to 79.71 km and ignites about 49.40 s later than the 0° trajectory. Its selected

ship position is 82.31 km farther downrange, and its separation load is 1.52 t higher. These differences describe the two optimized trajectories with individually selected ship positions.

At 30°, the coast-first trial reaches the upper boundary of the implemented atmosphere about 22 s after separation, while the booster is still climbing. The accepted feedback trajectory starts a braking burn at separation. The burn lasts about 32.13 s, limiting the subsequent apex to 85.90 km. Of the 67.00 t loaded at separation, approximately 28.25 t is burned during ascent braking, 33.41 t during the landing burn, and 5.34 t remains at contact. The 67.00 t load gives a feasible upper bound for this case within the adopted model. Lower-inventory candidates failed the replay checks for atmospheric coverage or touchdown, so the available runs establish feasibility at 67.00 t and leave the minimum unresolved.

7 Sensitivity and Uncertainty Analysis

7.1 Initial Flight-Path Angle and Atmospheric Coverage

We first use an unpowered ascent to determine whether the selected flight-path angles stay within the range of the implemented atmosphere. The initial body pitch follows the flight-path angle in each case. At 0° the booster starts descending from 70.0 km. At 15° the simulated ascent reaches approximately 79.71 km. At 30° an integration stage exceeds the atmosphere ceiling near 86 km while the vehicle is still climbing. The last supported step is at 85.99 km, about 22.0 s after separation. The 30° result identifies a model coverage limit for a coast-first trajectory. Evaluating that trajectory through its apex requires a higher-altitude atmosphere model.

7.2 Wind, Initial Attitude, and Fixed-Ship Capture

We replay the selected one-ignition schedule from separation with the ship fixed at 278.310 km downrange and the initial inventory fixed at 26.228 t. The ignition time, throttle, and gimbal commands are held at their nominal histories, so the replay does not correct for perturbations. We change one input at a time. Wind is uniform and positive downrange. Pitch offsets are applied to the body alone, leaving the initial flight-path angle at 15°. Each replay runs until first contact with the deck plane or the end of the P2 replay window, which extends 0.5 s beyond the last control knot.

The nominal replay passes all P0 touchdown criteria. In Table 3, the -10 m/s wind case and the $-2°$ pitch case end with the lowest point of the body still 9.6 m and 46.8 m above the deck plane, respectively. Each of the four perturbed trajectories that reaches the deck plane exceeds the 5 m position tolerance and both speed limits. The $+1°$/s pitch-rate case reaches the deck plane within 2° of upright but still exceeds the position, speed, and pitch-rate limits. Thus the conditions for a controlled landing must be met together. These deterministic cases characterize the nominal commands without feedback correction. Estimating recovery under perturbations would require replaying a guidance law that adjusts the trajectory toward the fixed ship.

Table 3: Full-flight replay with the nominal P2 commands and fixed ship position. The signed position error is measured from the ship. Positive vertical velocity points upward. Contact quantities are shown only when the booster reaches the deck plane.

Input change	Outcome	Δx (m)	v_x (m/s)	v_h (m/s)
Nominal	Within limits	0.0	−0.01	−1.53
Wind −10 m/s	Schedule ended	–	–	–
Wind +10 m/s	Outside limits	+378.3	9.31	−14.80
Pitch −2°	Schedule ended	–	–	–
Pitch +2°	Outside limits	+139.0	25.97	−25.16
Pitch rate −1°/s	Outside limits	−1061.7	151.71	−71.68
Pitch rate +1°/s	Outside limits	+34.1	106.40	−72.73

7.3 Separation-Time Replanning at a Fixed Inventory

We next use the initial state and steady wind as known inputs at separation and compute a new schedule for each case. The ship remains at the reference location. We select a common separation inventory of 30 t and retain the 5 t minimum usable reserve. A coast prediction supplies a state at 25 km. From that state, the optimizer chooses the remaining coast duration and the throttle and gimbal histories, with fixed initial mass and fixed ship position. The resulting commands are replayed from separation, including the preceding coast. This procedure uses the flight time before ignition to prepare a case-specific schedule.

The reference case and ten one-factor perturbations satisfy all P0 touchdown criteria at replay steps of 0.005 s and 0.0025 s. Table 4 uses the finer replay. The largest position error among these cases is approximately 0.026 m, and the largest downward contact speed is approximately 1.584 m/s. Each case retains more than 7.16 t. With the same 30 t inventory but the original P2 commands, all eleven cases fail at least one touchdown criterion.

Table 4: Full-flight replay after case-specific planning at separation. The ship position and 30 t initial inventory are common to all cases. Each row satisfies the P0 touchdown criteria.

Input change	Retained (t)	$\lvert v_x \rvert$ (m/s)	$\lvert v_h \rvert$ (m/s)
Nominal	7.882	0.03	1.58
Wind −1 m/s	7.939	0.00	1.49
Wind +1 m/s	7.793	0.03	1.42
Pitch −0.1°	7.866	0.03	1.42
Pitch +0.1°	7.861	0.02	1.43
Wind −10 m/s	8.497	0.00	1.51
Wind +10 m/s	7.165	0.02	1.45
Pitch −2°	7.951	0.00	1.51
Pitch +2°	7.809	0.02	1.54
Pitch rate −1°/s	7.434	0.02	1.56
Pitch rate +1°/s	7.551	0.00	1.49

The common 30 t inventory is a trial design choice, 3.772 t above the nominal P2

inventory. These runs establish feasibility for the individual modeled cases. The minimum common inventory and combined disturbances remain open questions. Planning takes approximately 14–43 s on the machine used here, with first ignition approximately 173–176 s after separation. These timings describe the recorded computations and leave room before ignition in these runs.

The calculation assumes exact initial-state information, known uniform wind, and the reference dynamics. Commands are then executed as a precomputed schedule. Wind changes or state errors arising later in flight require further estimation and guidance updates. Actuator rate limits, sensing and computation delays, three-dimensional motion, Earth curvature and rotation, and structural and thermal limits also remain to be incorporated before operational safety can be assessed.

7.4 Error in the Wind Used for Planning

We hold the revised 30 t nominal schedule fixed and change the actual uniform wind while the wind used to plan the schedule remains zero. We examine thirteen wind speeds between -1 and $+1$ m/s, with more closely spaced points near zero. The event types and lists of violated touchdown limits agree for replay steps of 0.005 s and 0.0025 s.

The tested points from -0.05 through $+0.05$ m/s satisfy the touchdown criteria. At $+0.05$ m/s, the downward contact speed is 1.993 m/s and the pitch-rate magnitude is 0.945°/s, close to their respective limits of 2 m/s and 1°/s. At $+0.1$ m/s, the position error remains within tolerance at 3.833 m, but the downward speed rises to 2.338 m/s and the pitch-rate magnitude to 1.608°/s. At -0.1 m/s, the command window ends with the lowest point of the body 0.571 m above the deck plane.

These results quantify the remaining sensitivity to wind information in the revised nominal schedule. The successful case-specific plans use the actual steady wind as an input. Continued adjustment to state deviations is needed to address errors in that input. The examined wind points provide a sampled comparison, with the precise feasibility boundary left for further analysis.

7.5 Tracking with Control Margins

We prepare a new nominal reference with throttle limited to 95% of maximum thrust and gimbal deflection limited to 9°. Ignition is advanced by 1 s relative to the preceding 30 t nominal schedule, to approximately 174.14 s after separation. The planned contact speed is reduced to 1 m/s downward. A penalty on changes between adjacent control nodes encourages smoother commands. The execution model retains the P0 actuator magnitude limits.

We linearize the dynamics along the powered reference and compute finite-horizon quadratic tracking gains before flight. During the burn, the controller reads the state every 20 ms, applies a correction to the reference throttle and gimbal commands, and holds those commands until the next update. Corrections are clipped to the permitted actuator range. The controller uses the nominal zero-wind model. Actual wind affects the simulated vehicle through the aerodynamic forces.

For normalized input $\mathbf{u} = (u_T, \delta/\delta_{max})$, the correction is

$$\mathbf{u}_{cmd} = \mathrm{clip}_{\mathcal{U}}\big[\mathbf{u}_{ref} - K(t)S^{-1}(\mathbf{y} - \mathbf{y}_{ref})\big], \qquad \mathcal{U} = [0.4, 1] \times [-1, 1]. \tag{25}$$

The seven diagonal scales in S are 100 m, 100 m, 30 m/s, 30 m/s, 1000 kg, 0.2 rad, and 0.2 rad/s, in the P1 state order. Pitch error is wrapped to its equivalent principal angle.

Numerical transition derivatives on a reference mesh of at most 0.1 s supply the state and input matrices for a backward Riccati recursion. Reference states use cubic Hermite interpolation, and the gain is held over each reference interval. The saved controller file specifies the quadratic weights and gains.

Table 5: Tracking results for unknown steady wind relative to the zero-wind reference. Each feedback case satisfies all P0 touchdown criteria at both integration step sizes. Values use the 0.0025 s replay.

Wind (m/s)	Position error (m)	Downward speed (m/s)	Retained (t)
−0.10	2.882	0.587	6.043
−0.05	1.422	0.585	6.041
0	0.038	0.582	6.038
+0.05	1.495	0.580	6.036
+0.10	2.953	0.578	6.034

The largest touchdown pitch-rate magnitude is approximately 0.740°/s. Peak commanded throttle and gimbal magnitude are approximately 95.9% and 9.09°, within the original capability limits. Holding the same new reference commands fixed passes only the zero-wind case among these five comparisons. The feedback cases consume approximately 23.96 t and retain more than 6.03 t from the common 30 t separation load.

These comparisons assume exact state measurements and instantaneous actuator response, with a fixed 20 ms control update period. Measurement noise, additional delay, actuator rate limits, changing wind, combined initial errors, and a broader wind range remain to be assessed. Ignition uses a prescribed nominal time.

7.6 Flight Loads and Aerothermal Exposure

We now separate the touchdown requirements from the environment encountered before contact. We evaluate the selected P2 trajectory and the five P3 feedback cases using the same P1 forces, with a 0.0025 s integration step and the existing 20 ms feedback period. The control schedules, initial inventories, and ship position are unchanged. Figure 4 compares the two nominal trajectories.

The dynamic pressure is $q = \rho U^2/2$, where U is the speed relative to the air [8]. The non-gravitational specific-force magnitude at the centre of mass is

$$n = \frac{\|\mathbf{F}_a + \mathbf{T}\|}{m g_0}. \tag{26}$$

This quantity differs from the magnitude of the ground-relative acceleration, which also includes gravity. Body-axis force components, integrated normal aerodynamic force, and external pitch moments provide additional information about loading. Neither n nor a resultant moment about the centre of mass determines the stress or internal bending moment at a structural section.

Peak dynamic pressure occurs during the unpowered descent, 3.23 s before ignition. At this point the air-relative speed is approximately 1180 m/s. The specific-force peak occurs later, during the burn, when the body axis makes an acute angle of approximately 18.39° with the air-relative velocity line. At that instant the normal aerodynamic force is approximately 9.535 MN, compared with 4.852 MN axially. Its magnitude subsequently

Table 6: Peak environment quantities for the nominal 30 t feedback trajectory. Altitude denotes centre-of-mass height. The energy-flux entry describes the incoming flow, not heat transferred to the surface.

Quantity	Time (s)	Altitude (km)	Peak
Flow kinetic-energy flux	168.980	11.019	382.31 MW/m^2
Dynamic pressure	170.905	9.553	304.315 kPa
Specific-force magnitude	181.577	3.400	8.132 g_0
Normal aerodynamic force	181.597	3.392	9.541 MN
Aerodynamic pitch moment	181.445	3.454	35.099 MN m

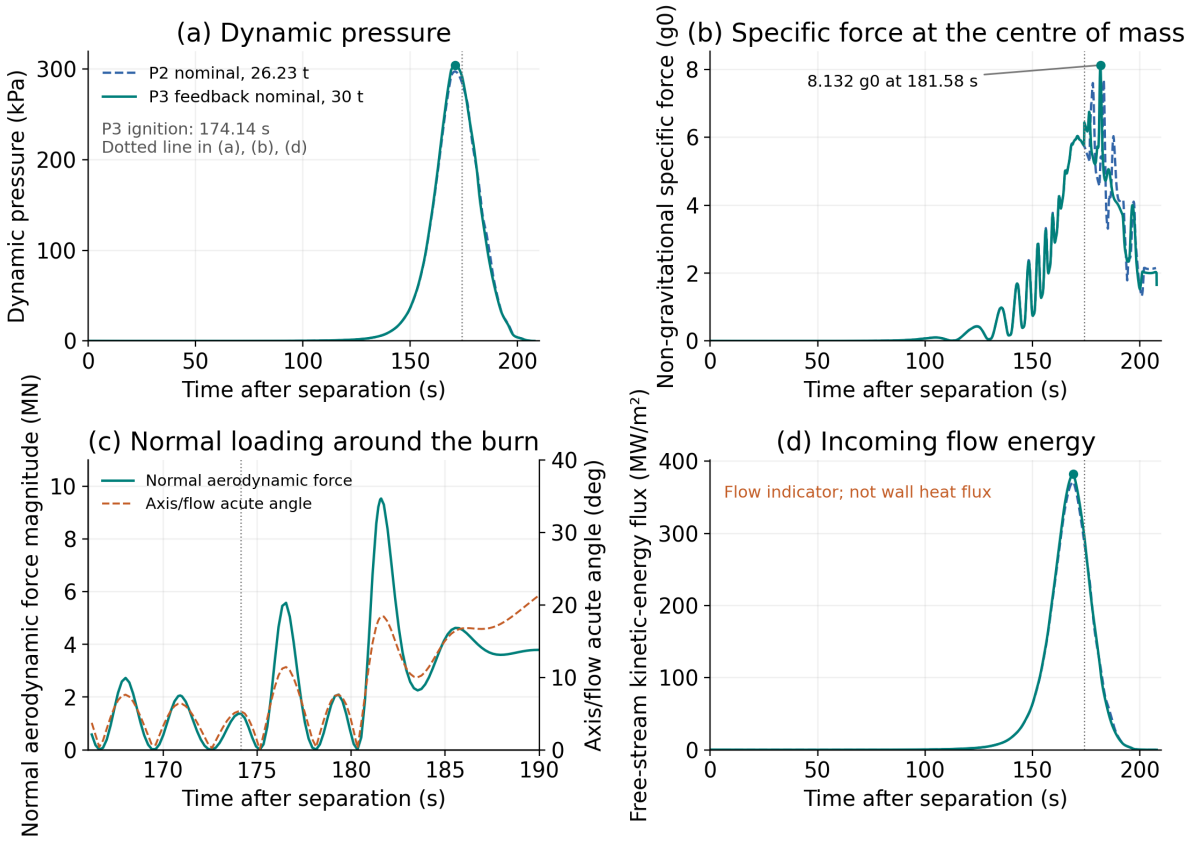


Figure 4: Nominal flight environments. Dotted lines mark P3 ignition. Panel (c) uses P3 alone and the acute body-axis/flow-line angle. Panel (d) is a flow-energy indicator, not wall heat flux. Inventory, reference planning, and feedback differ between P2 and P3.

reaches the value in Table 6. The crossflow strip model includes local velocity due to rotation, so the reported normal force is not inferred from the centre-of-mass angle alone.

At the specific-force peak, aerodynamic and thrust magnitudes divided by mg_0 are approximately 6.972 and 1.761, respectively. These magnitudes cannot be added as scalars; their vector sum gives 8.132. In body coordinates the combined components are approximately 4.901 g_0 axially and $-6.489\,g_0$ normally. The strong normal loading explains why a lower dynamic pressure during the burn can coincide with a higher specific force than at max-q.

The P2 nominal peaks are 297.334 kPa and 7.703 g_0. The P3 nominal values are higher by approximately 2.35% and 5.56%, respectively. Improved touchdown recovery therefore

comes with increased peak requirements in this comparison; it does not establish reduced structural loading. This comparison changes inventory and reference planning together with feedback, and does not isolate the effect of feedback alone. Across the five small wind-error cases, peak dynamic pressure ranges from 304.302 to 304.327 kPa and peak specific force from 8.1311 to 8.1325 g_0 (rounded outward). This narrow sampled variation is not a result for broader wind or combined errors.

For thermal exposure, we record the free-stream kinetic-energy flux per area normal to the flow,

$$\Phi_k = \tfrac{1}{2}\rho U^3 = qU. \tag{27}$$

This flow quantity identifies a period when dense, fast air carries substantial kinetic energy. It peaks before both ignition and max-q. It is distinct from the wall heat flux and its time integral is not the heat absorbed by the booster. Surface heating depends on local flow, geometry, wall conditions, and heat transfer [9]. The Sutton–Graves study concerns stagnation-point heating of axisymmetric blunt bodies [10]. Applying such a relation here requires a justified effective leading-surface radius and flow regime; the cylinder diameter alone does not supply these inputs for every attitude. We therefore report Φ_k as an exposure indicator and leave wall heat flux, surface temperature, and thermal-protection thickness undetermined.

The five trajectories meet the adopted touchdown constraints while imposing the load requirements above. These magnitudes inherit the constant-coefficient aerodynamics and rigid-body approximations. Structural and thermal safety remain undetermined without allowables, structural geometry and material properties, and local heating and material-response models. Both max-q and peak flow energy precede ignition. Reducing them requires changing the preceding descent or introducing earlier braking or attitude control; the present burn-only feedback cannot change an already encountered peak. Such changes require a new propellant estimate.

7.7 Whole-Flight Parameters and Reserve

We first examine effective engine efficiency using the P0 scenario points. A recovery site selected separately for each design differs from an already positioned ship that must remain fixed across disturbances.

7.7.1 Effective Specific Impulse and the Separation Budget

We compare the P0 effective specific impulses of 300, 330, and 360 s while keeping maximum thrust, minimum throttle, gimbal limits, aerodynamic parameters, initial direction, and the 5 t reserve requirement unchanged. The oxygen-to-methane mass ratio remains 3.5. This is an effective-parameter comparison, rather than a thermochemically coupled engine redesign. Each value is treated as known when planning the trajectory.

For each changed value, we reoptimize the separation inventory, phase durations, throttle, and gimbal history using the P2 single-ignition family and a 0.55 s target shooting interval. Each design selects its own ship position, which is then fixed during full-flight replay. The 330 s row retains the selected P2 solution. All three rows satisfy the P0 touchdown criteria at integration steps of 0.005 s and 0.0025 s. Table 7 uses the latter. These are accepted local candidates; the search does not establish global minima.

Relative to 330 s, accepted inventory rises by 2.552 t (9.73%) at 300 s and falls by 2.029 t (7.74%) at 360 s. Figure 5 separates consumption from the nearly constant retained

Table 7: Effective-Isp scenario budgets in tonnes. Separation inventory includes the retained amount, which exceeds the common 5 t requirement. Methane and oxygen are components of the separation inventory. Values are rounded independently.

I_{sp} (s)	Inventory	Consumed	Retained	Methane	Oxygen
300	28.781	23.701	5.079	6.396	22.385
330	26.228	21.150	5.078	5.828	20.400
360	24.199	19.124	5.075	5.378	18.821

inventory. The examined separation-load range is 24.199–28.781 t, including 5.378–6.396 t of methane. This is a deterministic scenario range, not a confidence interval or a universal requirement for the unspecified booster.

At a prescribed thrust, $\dot{m} = -T/(g_0 I_{sp})$ links efficiency to consumption. However, scaling the reference consumed mass by $330/I_{sp}$ and retaining its 5.078 t endpoint inventory would predict total loads of 28.343 t at 300 s and 24.466 t at 360 s. These differ from the reoptimized values. Initial inventory affects the mass throughout the coast, while the optimized controls and durations also change. The total thrust impulse therefore need not remain that of the reference trajectory.

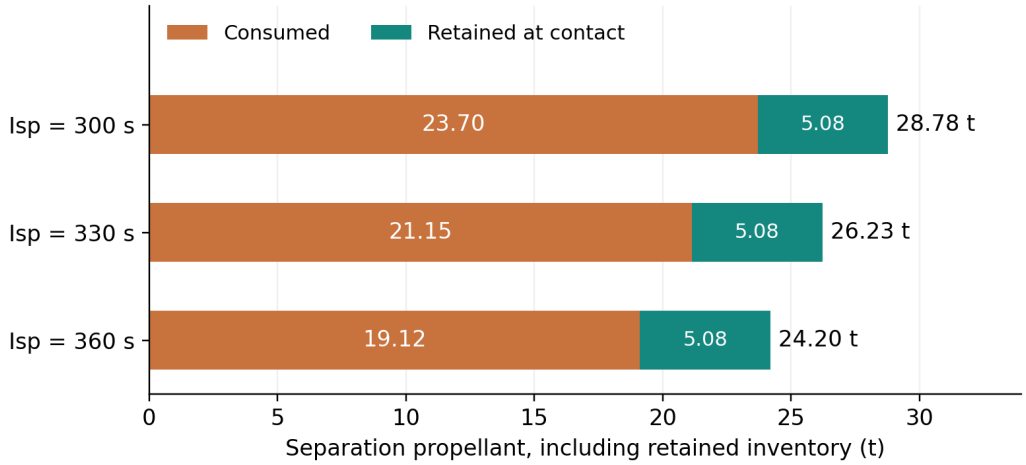


Figure 5: Separation inventory split into consumed and retained propellant for the three known-efficiency designs. Each design has its own optimized commands and selected recovery site.

The selected ship positions for 300, 330, and 360 s are 278.455, 278.310, and 278.260 km downrange, respectively. Ignition occurs at approximately 176.27, 176.68, and 176.70 s. The changed plans are thus separate design cases, rather than a demonstration that one command history recovers unknown engine errors at a fixed ship. Their inventories also differ from the 30 t feedback trials of the preceding subsections.

The corresponding peak dynamic pressures are approximately 302.062, 297.334, and 293.561 kPa, and peak specific forces are 7.835, 7.703, and 7.636 g_0. These values retain the structural and thermal limitations of Section 7.6. The budgets meet the adopted touchdown conditions without additional structural or thermal constraints. Thrust, aerodynamic, and reserve variations remain separate calculations.

Draft plan: Continue with separately reoptimized thrust, aerodynamic, and reserve scenarios. Reserve changes vehicle mass as well as the amount retained. Report the governing touchdown or flight limit for each case.

8 Model Evaluation

8.1 Overview and Comparison of Model Levels

8.2 Verification and Validation

8.3 P1 Internal Verification and Numerical Convergence

The verification suite tests implementation consistency and known limiting solutions; no flight-data validation is claimed. Standard-atmosphere base pressures at 0, 11, 20, 32, and 47 km geopotential altitude agree with rounded reference values within 4.8×10^{-6} relative error. Out-of-range atmospheric queries are rejected. A vertical vacuum coast conserves the modeled specific mechanical energy to 7.5×10^{-9} J/kg over 20 s. Constant-thrust mass history and horizontal ideal-rocket velocity agree with their analytic expressions to 7.6×10^{-10} kg and 2.9×10^{-14} m/s, respectively, in the chosen tests. These small residuals apply to those tests and do not measure real-world accuracy.

Pure axial and pure crossflow force limits, static pressure-center torque, torque-free rotation, damping sign, gimbal sign, and air-relative dissipativity are checked. For pure centered rotation, Eq. (9) provides an independent integral. Midpoint-strip relative errors are 2%, 0.5%, 0.125%, 0.03125%, and 0.0078125% for 10, 20, 40, 80, and 160 strips. Thus the default 20-strip rotational torque has a 0.5% quadrature error in this limiting case; this is distinct from uncertainty in the aerodynamic coefficients.

A four-second smooth unpowered trajectory is compared with a 0.00125 s reference using the same equations. Position differences decrease from 9.75×10^{-7} m at 0.08 s to 2.76×10^{-10} m at 0.01 s. Since force laws contain absolute values and feedback contains clipping, fourth-order behavior is not assumed across every nonsmooth transition. Figure 6 separates smooth integration, strip quadrature, and touchdown-event checks. Between the final two steps, consumption changes by 0.00384 kg and contact

Table 8: Local terminal-event convergence. All rows satisfy the selected speed, attitude, and reserve criteria; this table does not test fixed-ship position capture.

Step (s)	Contact time (s)	v_h (m/s)	Consumed (kg)
0.080	35.975945	-1.502940	22679.544
0.040	35.975286	-1.503756	22679.191
0.020	35.975097	-1.502978	22679.139
0.010	35.975063	-1.502986	22679.123
0.005	35.975047	-1.502909	22679.119

time by 1.60×10^{-5} s. Event speed differences are not monotonic, so no fourth-order contact-convergence claim is made. The fuel-depletion guard is verified to avoid dry-mass overshoot, while exact depletion-time localization remains outside this conservative guard's contract. The reference implementation and P1 checks are reproducible with

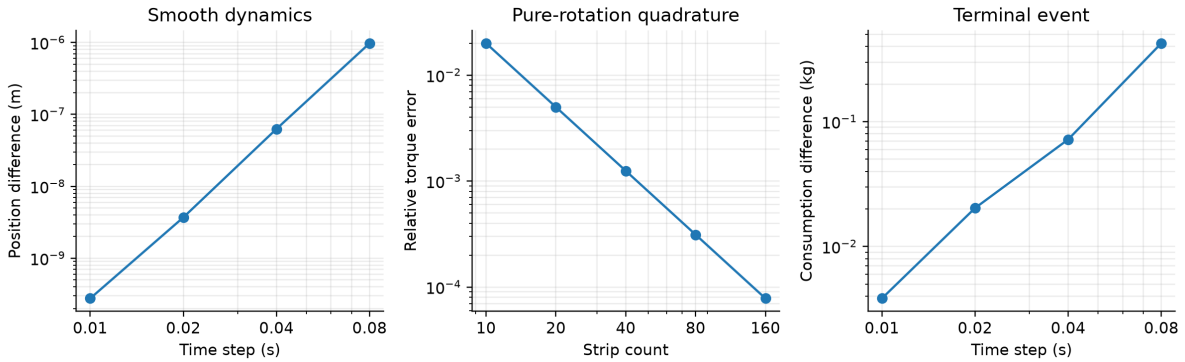


Figure 6: Independent numerical error sources. Left: smooth-trajectory position difference relative to a finer same-model integration. Middle: strip torque error against the analytic integral. Right: contact-consumption difference relative to the 0.005 s run.

`node src/verify-p1.mjs`. The remaining limitations are model-form uncertainty, actuator idealization, atmosphere range, local-plane/rotation approximations, and absent structural/thermal allowables. Whole-flight feasibility and fuel optimization are delegated to the next stage.

8.4 Strengths and Weaknesses

8.5 Discussion of Sources of Error

8.5.1 Simplified Physical Forces and Attitude Dynamics

8.5.2 Numerical Computation Error

8.5.3 Parameter and Operational Uncertainty

9 Further Discussion

9.1 Attitude Stability and Vertical Touchdown

The local pitch controller brings the body upright under the tested conditions, but the case with an initial pitch-rate error illustrates that attitude recovery can occur alongside a position miss. A full recovery trajectory must satisfy the position, velocity, tilt, and pitch-rate limits at the same contact event. Control authority also depends on gimbal saturation and whether the engine is on.

9.2 Structural Loads and Aerothermal Constraints

Section 7.6 identifies distinct pre-ignition pressure and flow-energy peaks and a later normal-load peak during the burn. The improved touchdown results do not remove these load requirements. A design with lower allowables would need a different descent trajectory and a revised propellant budget; the present contract does not provide those allowables.

9.3 Ship Placement, Wind, and Deck Motion

Downrange placement can avoid a return maneuver. Once the ship is positioned, the guidance must target its location and meet the speed and attitude limits relative to its deck. The present stationary, horizontal deck sets the reference for the fixed-site examples. A moving deck would alter the target position, relative velocity, and contact orientation.

9.4 Conclusions and Practical Recommendations

Draft plan: Answer the separation-fuel question with assumptions, units, uncertainty, and feasibility limits.

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A Reproducibility and Source Code

From the repository root, run `node src/run-planar.mjs` to reproduce the demonstration trajectories and `node src/verify-p1.mjs` to reproduce the P1 numerical comparisons.

Run `node src/p3-scenarios.mjs` to reproduce the fixed-ship and atmosphere coverage cases. The command reads the reference contract and the selected P2 command history, then writes `results/p3-scenarios.json`. The following commands compute the reference corrective schedule and rebuild the full-flight comparison from the saved schedules.

```
python src/p3-replan.py --case nominal
python src/p3-replan.py --replay-all
```

The schedules are stored in `results/p3/replanned/`. The planner uses the dependencies in `requirements-p2.txt`. Run `python src/p3-tracking.py` to prepare the control-margin reference and regenerate the tracking gains and two-step-size comparison in `results/p3/tracking/`. Run `node src/p3-exposure.mjs` followed by `python src/plot-p3-exposure.py` to generate the full-flight environment records in `results/p3/safety/` and their figure. The incoming flow-energy indicator is not a wall-heating calculation. Run `python src/p3-propulsion.py --isp 330`, followed by the same command with `--isp 300` and `--isp 360`, to reproduce the effective-efficiency scenarios. Run `python src/plot-p3-propulsion.py` to generate their budget figure. The saved candidates and scenario budgets are in `results/p3/propulsion/`.
Draft plan: Add the full-flight solver and final figure generation commands and versions.

B Supplementary Derivations and Results

Draft plan: Include dimensional checks, convergence tables, and additional scenario results.

Report on the Use of AI

AI assistance was used to organize the manuscript, develop the reference dynamics and verification scripts, derive model equations, and generate P1 figures and draft text. It was also used to implement the P2 search, optimization, and replay, generate its figure, and draft the landing-strategy section. OpenAI Codex also assisted with the sensitivity calculations, corrective planning, tracking-controller implementation, flight-environment and propulsion-scenario analysis and figures, and manuscript integration.
Draft plan: Complete the disclosure with the tools used, exact prompts, and complete outputs required by the competition rules.